

Representations of Lie Groups

A Mathematical Dissection of Quantum Spin

Part I: Mathematical Foundations

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Topological Spaces

Definition (topological space)

A **topological space** is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfies:

- ① (Trivial bounds) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- ② (Arbitrary unions) $\forall \{U_\alpha\}_{\alpha \in \mathcal{I}} \subseteq \mathcal{T} : \bigcup_{\alpha \in \mathcal{I}} U_\alpha \in \mathcal{T}$.
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Key constructions. Product topology, subspace topology, metric topologies (e.g. the Euclidean topology on $\mathbb{R}^n$).

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Why Hausdorff + second-countable? These are technical conditions that prevent pathological spaces. They guarantee that partitions of unity exist.

Definition (smooth manifold)

A **smooth manifold** is a triple $(M, \mathcal{T}, \mathcal{A})$ where:

- ① $(M, \mathcal{T})$ is a topological manifold.
- ② $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{I}}$ is a **smooth atlas**: each $(U_\alpha, \varphi_\alpha)$ is a chart, the charts cover M , and transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ are C^∞ .
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Key idea. Charts are local coordinate systems. Smoothness means that when you switch between coordinate systems, the change-of-coordinates map is infinitely differentiable.

Smooth Maps and Diffeomorphisms

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Example. $\mathbb{R}^n$ is diffeomorphic to the open unit ball $B_1(0)$. S^n is *not* diffeomorphic to $\mathbb{R}^n$ (compact vs. non-compact).

Tangent Spaces via Derivations

Motivation

How do we define tangent vectors to a smooth manifold without embedding it in $\mathbb{R}^N$? Answer: **derivations** — linear maps that satisfy the Leibniz rule.

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Let M be a smooth manifold, $p \in M$. The **stalk** $C_p^\infty M$ is the set of germs of smooth functions at p . The **tangent space** at p is

$$T_p M \stackrel{\text{df}}{=} \{v : C_p^\infty M \rightarrow \mathbb{R} : v \text{ is a derivation at } p\}.$$

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Why derivations? For $\mathbb{R}^n$, a tangent vector v acts on functions by the directional derivative: $v(f) = \sum_i v^i \frac{\partial f}{\partial x^i} \Big|_p$. This algebraic characterisation generalises to any manifold.

The Differential

Definition (differential)

Let $F : M \rightarrow N$ be smooth, $p \in M$. The **differential of F at p** is the linear map $dF_p : T_p M \rightarrow T_{F(p)} N$ defined by

$$\forall v \in T_p M, \forall [f]_{F(p)} \in C_{F(p)}^\infty N : \quad [(dF_p(v))]([f]_{F(p)}) \stackrel{\text{df}}{=} v([f \circ F]_p).$$

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Interpretation. The differential is the “best linear approximation” to a smooth map, generalising the Jacobian matrix from multivariable calculus.

Groups, Fields, and Vector Spaces

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A set G with an associative binary operation μ , an identity e , and inverses ι .
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Why these matter. Lie groups combine the smooth structure of manifolds with the algebraic structure of groups. Lie algebras are vector spaces with an extra bracket operation.

Definition ($\mathbb{F}$ -algebra)

A vector space $\mathcal{A}$ over $\mathbb{F}$ together with a bilinear multiplication $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$.
(Examples: $\mathbb{R}^{n,n}$ with matrix multiplication, $C^\infty(M)$ with pointwise multiplication.)

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An $\mathbb{F}$ -vector space $\mathfrak{g}$ with a bilinear bracket $[\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying:

- ① (Skew-symmetry) $[Y, X] = -[X, Y]$
- ② (Jacobi identity) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$

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Key example. $(\mathbb{R}^3, \times)$ — the cross product makes $\mathbb{R}^3$ a Lie algebra. $[v, w] = v \times w$ satisfies skew-symmetry and Jacobi.

Lie Groups

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A **Lie group** is a smooth manifold $(G, \mathcal{T}, \mathcal{A})$ that is also a group $(G, \{\mu, \iota, e\})$ such that:

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Examples.

- $(\mathbb{R}^n, +)$ — the translation group.
- $\mathrm{GL}(n, \mathbb{R})$ — invertible $n \times n$ real matrices.
- $\mathrm{SO}(n)$ — rotations in $\mathbb{R}^n$.
- $\mathrm{SU}(n)$ — unitary transformations with determinant 1.

Left Translation and Left-Invariant Vector Fields

Definition (left translation)

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Geometric meaning. A left-invariant vector field is completely determined by its value at the identity e . Translating the vector at e by L_g gives the vector at any point g .

The Lie Algebra of a Lie Group

Theorem (tangent space at identity is a Lie algebra)

Let G be a Lie group. Define $\mathfrak{g} = T_e G$ (tangent space at identity). For each $v \in \mathfrak{g}$, there exists a **unique** left-invariant vector field X^v with $X^v|_e = v$. Define the bracket on $\mathfrak{g}$ by

$$[v, w]_{\mathfrak{g}} \stackrel{\text{df}}{=} [X^v, X^w]|_e.$$

Then $(\mathfrak{g}, [\bullet, \bullet]_{\mathfrak{g}})$ is a Lie $\mathbb{R}$ -algebra with $\dim(\mathfrak{g}) = \dim(G)$.

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Remark. We denote the Lie algebra of a Lie group G by $\text{Lie}(G)$.

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Let G be a Lie group and $H \subseteq G$ a closed subgroup. Then H admits a unique smooth structure making it a Lie group such that the inclusion $H \hookrightarrow G$ is a smooth embedding.

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Example. $SO(n)$, $SU(n)$, $SL(n, \mathbb{R})$, and $Sp(2n)$ are all closed subgroups of $GL(N, \mathbb{F})$ and therefore Lie groups.

The General Linear Group $GL(n, \mathbb{F})$

Example (the general linear group)

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- Its Lie algebra is $\mathfrak{gl}(n, \mathbb{F}) = \mathbb{F}^{n,n}$ (all $n \times n$ matrices) with the commutator bracket:

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- Most matrix Lie groups arise as closed subgroups of $GL(n, \mathbb{F})$. Their Lie algebras are subspaces of $\mathfrak{gl}(n, \mathbb{F})$ closed under the commutator.

The Special Orthogonal Group $SO(3)$

Example (the special orthogonal group — rigid rotations)

$$SO(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1 \};$$

$$\mathfrak{so}(3) = T_{\mathbf{I}_3} SO(3) = \{ \mathbf{R} \in \mathbb{R}^{3,3} : \mathbf{R}^T = -\mathbf{R} \}.$$

$\dim(\mathfrak{so}(3)) = \dim(SO(3)) = 3$, with basis $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$:

$$\mathbf{L}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad \mathbf{L}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

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Remark. $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$ — the same structure as the cross product on $\mathbb{R}^3$.

The Special Unitary Group $SU(2)$

Example (the special unitary group)

$$SU(2) = \left\{ \mathbf{U} \in \mathbb{C}^{2,2} : \mathbf{U}^\dagger \mathbf{U} = \mathbf{I}_2, \det(\mathbf{U}) = 1 \right\};$$

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$\dim(\mathfrak{su}(2)) = \dim(SU(2)) = 3$, with basis $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk} \mathbf{T}_k$:

$$\mathbf{T}_1 = \frac{1}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \mathbf{T}_2 = \frac{1}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{T}_3 = \frac{1}{2} \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}.$$

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Remark. $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk} \mathbf{T}_k$ — **identical** to $\mathfrak{so}(3)$. This is the first hint of the isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

The Double Cover $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$

Construction

For $\mathbf{U} \in \mathrm{SU}(2)$, define $\Phi(\mathbf{U}) : \mathfrak{su}(2) \rightarrow \mathfrak{su}(2)$ by

$$[\Phi(\mathbf{U})](\mathbf{X}) \stackrel{\mathrm{df}}{=} \mathbf{U}\mathbf{X}\mathbf{U}^\dagger.$$

Using the basis $\beta = (\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3)$, identify $\mathfrak{su}(2) \cong \mathbb{R}^3$. Then the matrix $[\Phi(\mathbf{U})]_\beta^\beta$ lies in $\mathrm{SO}(3)$.

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Theorem (double cover $\text{SU}(2) \rightarrow \text{SO}(3)$)

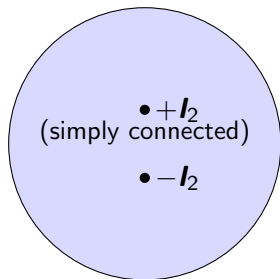
$\Phi : \text{SU}(2) \rightarrow \text{SO}(3)$ is:

- 1 A smooth group homomorphism: $\Phi(\mathbf{UV}) = \Phi(\mathbf{U}) \circ \Phi(\mathbf{V})$.
- 2 Surjective.
- 3 Has kernel $\ker(\Phi) = \{\pm \mathbf{I}_2\}$ (two elements).

Thus $\text{SU}(2)$ is a **double cover** of $\text{SO}(3)$.

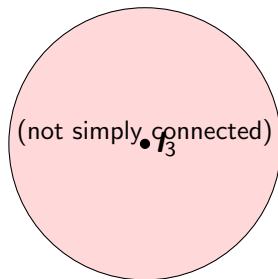
The Double Cover: Geometric Picture

$$\mathrm{SU}(2) \cong S^3$$

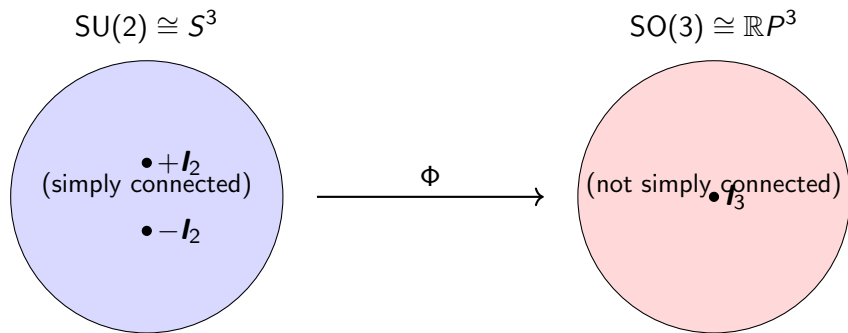


$$\xrightarrow{\Phi}$$

$$\mathrm{SO}(3) \cong \mathbb{R}P^3$$



The Double Cover: Geometric Picture



Why two-to-one? Both $\mathbf{U}$ and $-\mathbf{U}$ give the same rotation. A rotation by angle θ about axis $\hat{\mathbf{n}}$ lifts to $\pm \cos(\theta/2)\mathbf{I}_2 \mp i \sin(\theta/2)(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})$. Increasing θ by 2π flips the sign.

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Define $\varphi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$ by $\varphi(\mathbf{L}_i) \stackrel{\text{df}}{=} \mathbf{T}_i$ (and extend $\mathbb{R}$ -linearly). Then φ is a Lie algebra isomorphism.

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Proof (essence)

- φ maps a basis to a basis $\implies$ linear isomorphism.
- Both satisfy $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k$ and $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk} \mathbf{T}_k$.
- Therefore $\varphi([\mathbf{L}_i, \mathbf{L}_j]) = \varphi(\varepsilon_{ijk} \mathbf{L}_k) = \varepsilon_{ijk} \mathbf{T}_k = [\mathbf{T}_i, \mathbf{T}_j] = [\varphi(\mathbf{L}_i), \varphi(\mathbf{L}_j)]$.

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Significance. Although $SU(2)$ and $SO(3)$ are different as groups (double cover vs. quotient), their Lie algebras are **identical**. Infinitesimal rotations are the same in both pictures.

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- 6 $SO(3)$ (rotations) and $SU(2)$ (unitary 2×2 matrices) are the central examples.
- 7 $SU(2)$ is a **double cover** of $SO(3)$. Their Lie algebras are **isomorphic**: $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

What Comes Next (Part II)

Representation theory

- Representations of Lie algebras and Lie groups.
- Complexification $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$ and the Cartan–Weyl basis (H, E, F) .
- Classification of irreducible representations of $\mathfrak{su}(2)$: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, a unique irrep of dimension $2j + 1$.
- Integration: every $\mathfrak{su}(2)$ -representation exponentiates to an $SU(2)$ -representation.
- Projective representations and the descent to $SO(3)$: integer $j \rightarrow$ ordinary representations, half-integer $j \rightarrow$ spinor (projective) representations.
- Physical spin operators and the 4π periodicity of spinors.